

[INICIAL] TCArg 2026 Contest #5**A. DZY Loves Chemistry**

1 second, 256 megabytes

DZY loves chemistry, and he enjoys mixing chemicals.

DZY has n chemicals, and m pairs of them will react. He wants to pour these chemicals into a test tube, and he needs to pour them in one by one, in any order.

Let's consider the danger of a test tube. Danger of an empty test tube is 1. And every time when DZY pours a chemical, if there are already one or more chemicals in the test tube that can react with it, the danger of the test tube will be multiplied by 2. Otherwise the danger remains as it is.

Find the maximum possible danger after pouring all the chemicals one by one in optimal order.

Input

The first line contains two space-separated integers n and m ($1 \leq n \leq 50$; $0 \leq m \leq \frac{n(n-1)}{2}$).

Each of the next m lines contains two space-separated integers x_i and y_i ($1 \leq x_i < y_i \leq n$). These integers mean that the chemical x_i will react with the chemical y_i . Each pair of chemicals will appear at most once in the input.

Consider all the chemicals numbered from 1 to n in some order.

Output

Print a single integer — the maximum possible danger.

input
1 0
output
1

input
2 1 1 2
output
2

input
3 2 1 2 2 3
output
4

In the first sample, there's only one way to pour, and the danger won't increase.

In the second sample, no matter we pour the 1st chemical first, or pour the 2nd chemical first, the answer is always 2.

In the third sample, there are four ways to achieve the maximum possible danger: 2-1-3, 2-3-1, 1-2-3 and 3-2-1 (that is the numbers of the chemicals in order of pouring).

B. Hotelier

1 second, 256 megabytes

Amugae has a hotel consisting of 10 rooms. The rooms are numbered from 0 to 9 from left to right.

The hotel has two entrances — one from the left end, and another from the right end. When a customer arrives to the hotel through the left entrance, they are assigned to an empty room closest to the left entrance. Similarly, when a customer arrives at the hotel through the right entrance, they are assigned to an empty room closest to the right entrance.

One day, Amugae lost the room assignment list. Thankfully Amugae's memory is perfect, and he remembers all of the customers: when a customer arrived, from which entrance, and when they left the hotel. Initially the hotel was empty. Write a program that recovers the room assignment list from Amugae's memory.

Input

The first line consists of an integer n ($1 \leq n \leq 10^5$), the number of events in Amugae's memory.

The second line consists of a string of length n describing the events in chronological order. Each character represents:

- 'L': A customer arrives from the left entrance.

- 'R': A customer arrives from the right entrance.
- '0', '1', ..., '9': The customer in room x (0, 1, ..., 9 respectively) leaves.

It is guaranteed that there is at least one empty room when a customer arrives, and there is a customer in the room x when x (0, 1, ..., 9) is given. Also, all the rooms are initially empty.

Output

In the only line, output the hotel room's assignment status, from room 0 to room 9. Represent an empty room as '0', and an occupied room as '1', without spaces.

input
8 LLRL1RL1
output
1010000011

input
9 L0L0LLRR9
output
1100000010

In the first example, hotel room's assignment status after each action is as follows.

- First of all, all rooms are empty. Assignment status is 0000000000.
- L: a customer arrives to the hotel through the left entrance. Assignment status is 1000000000.
- L: one more customer from the left entrance. Assignment status is 1100000000.
- R: one more customer from the right entrance. Assignment status is 1100000001.
- L: one more customer from the left entrance. Assignment status is 1110000001.
- 1: the customer in room 1 leaves. Assignment status is 1010000001.
- R: one more customer from the right entrance. Assignment status is 1010000011.
- L: one more customer from the left entrance. Assignment status is 1110000011.
- 1: the customer in room 1 leaves. Assignment status is 1010000011.

So after all, hotel room's final assignment status is 1010000011.

In the second example, hotel room's assignment status after each action is as follows.

- L: a customer arrives to the hotel through the left entrance. Assignment status is 1000000000.
- 0: the customer in room 0 leaves. Assignment status is 0000000000.
- L: a customer arrives to the hotel through the left entrance. Assignment status is 1000000000 again.
- 0: the customer in room 0 leaves. Assignment status is 0000000000.
- L: a customer arrives to the hotel through the left entrance. Assignment status is 1000000000.
- L: one more customer from the left entrance. Assignment status is 1100000000.
- R: one more customer from the right entrance. Assignment status is 1100000001.
- R: one more customer from the right entrance. Assignment status is 1100000011.
- 9: the customer in room 9 leaves. Assignment status is 1100000010.

So after all, hotel room's final assignment status is 1100000010.

C. Bear and Poker

2 seconds, 256 megabytes

Limak is an old brown bear. He often plays poker with his friends. Today they went to a casino. There are n players (including Limak himself) and right now all of them have bids on the table. i -th of them has bid with size a_i dollars.

Each player can double his bid any number of times and triple his bid any number of times. The casino has a great jackpot for making all bids equal. Is it possible that Limak and his friends will win a jackpot?

Input

First line of input contains an integer n ($2 \leq n \leq 10^5$), the number of players.

The second line contains n integer numbers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the bids of players.

Output

Print "Yes" (without the quotes) if players can make their bids become equal, or "No" otherwise.

input
4 75 150 75 50
output
Yes

input
3 100 150 250
output
No

In the first sample test first and third players should double their bids twice, second player should double his bid once and fourth player should both double and triple his bid.

It can be shown that in the second sample test there is no way to make all bids equal.

D. Best Subsegment

1 second, 256 megabytes

You are given array $a_1, a_2, \dots, a_n$. Find the subsegment $a_l, a_{l+1}, \dots, a_r$ ($1 \leq l \leq r \leq n$) with maximum arithmetic mean $\frac{1}{r-l+1} \sum_{i=l}^r a_i$ (in floating-point numbers, i.e. without any rounding).

If there are many such subsegments find the **longest** one.

Input

The first line contains single integer n ($1 \leq n \leq 10^5$) — length of the array a .

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$) — the array a .

Output

Print the single integer — the length of the longest subsegment with maximum possible arithmetic mean.

input
5 6 1 6 6 0
output
2

The subsegment $[3, 4]$ is the longest among all subsegments with maximum arithmetic mean.

E. Suffix Structures

1 second, 256 megabytes

Bizon the Champion isn't just a bison. He also is a favorite of the "Bizons" team.

At a competition the "Bizons" got the following problem: "You are given two distinct words (strings of English letters), s and t . You need to transform word s into word t ". The task looked simple to the guys because they know the suffix data structures well. Bizon Senior loves suffix automaton. By applying it once to a string, he can remove from this string any single character. Bizon Middle knows suffix array well. By applying it once to a string, he can swap any two characters of this string. The guys do not know anything about the suffix tree, but it can help them do much more.

Bizon the Champion wonders whether the "Bizons" can solve the problem. Perhaps, the solution do not require both data structures. Find out whether the guys can solve the problem and if they can, how do they do it? Can they solve it either only with use of suffix automaton or only with use of suffix array or they need both structures? Note that any structure may be used an unlimited number of times, the structures may be used in any order.

Input

The first line contains a non-empty word s . The second line contains a non-empty word t . Words s and t are different. Each word consists only of lowercase English letters. Each word contains at most 100 letters.

Output

In the single line print the answer to the problem. Print "need tree" (without the quotes) if word s cannot be transformed into word t even with use of both suffix array and suffix automaton. Print "automaton" (without the quotes) if you need only the suffix automaton to solve the problem. Print "array" (without the quotes) if you need only the suffix array to solve the problem. Print "both" (without the quotes), if you need both data structures to solve the problem.

It's guaranteed that if you can solve the problem only with use of suffix array, then it is impossible to solve it only with use of suffix automaton. This is also true for suffix automaton.

input
automaton tomat
output
automaton

input
array arary
output
array

input
both hot
output
both

input
need tree
output
need tree

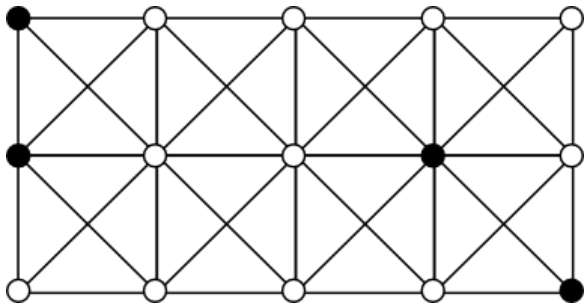
In the third sample you can act like that: first transform "both" into "oth" by removing the first character using the suffix automaton and then make two swaps of the string using the suffix array and get "hot".

F. Special Grid

4 seconds, 256 megabytes

You are given an $n \times m$ grid, some of its nodes are black, the others are white. Moreover, it's not an ordinary grid — each unit square of the grid has painted diagonals.

The figure below is an example of such grid of size 3×5 . Four nodes of this grid are black, the other 11 nodes are white.



Your task is to count the number of such triangles on the given grid that:

- the corners match the white nodes, and the area is positive;
- all sides go along the grid lines (horizontal, vertical or diagonal);
- no side contains black nodes.

Input
The first line contains two integers n and m ($2 \leq n, m \leq 400$). Each of the following n lines contain m characters (zeros and ones) — the description of the grid. If the j -th character in the i -th line equals zero, then the node on the i -th horizontal line and on the j -th vertical line is painted white. Otherwise, the node is painted black.

The horizontal lines are numbered starting from one from top to bottom, the vertical lines are numbered starting from one from left to right.

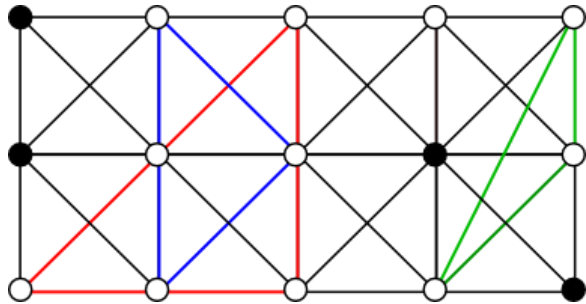
Output
Print a single integer — the number of required triangles.

input
3 5 10000 10010 00001
output
20

input
2 2 00 00
output
4

input
2 2 11 11
output
0

The figure below shows red and blue triangles. They are the examples of the required triangles in the first sample. One of the invalid triangles is painted green. It is invalid because not all sides go along the grid lines.



G. Gerald's Hexagon

2 s., 256 MB

Gerald got a very curious hexagon for his birthday. The boy found out that all the angles of the hexagon are equal to 120° . Then he measured the length of its sides, and found that each of them is equal to an integer number of centimeters. There the properties of the hexagon ended and Gerald decided to draw on it.

He painted a few lines, parallel to the sides of the hexagon. The lines split the hexagon into regular triangles with sides of 1 centimeter. Now Gerald wonders how many triangles he has got. But there were so many of them that Gerald lost the track of his counting. Help the boy count the triangles.

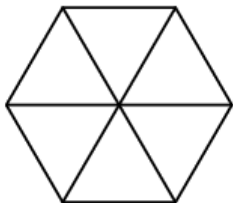
Input
The first and the single line of the input contains 6 space-separated integers a_1, a_2, a_3, a_4, a_5 and a_6 ($1 \leq a_i \leq 1000$) — the lengths of the sides of the hexagon in centimeters in clockwise order. It is guaranteed that an hexagon with the indicated properties and exactly such sides exists.

Output
Print a single integer — the number of triangles with the sides of one 1 centimeter, into which the hexagon is split.

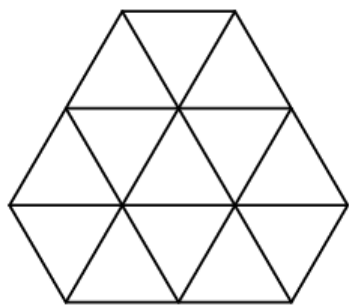
input
1 1 1 1 1 1
output
6

input
1 2 1 2 1 2
output
13

This is what Gerald's hexagon looks like in the first sample:



And that's what it looks like in the second sample:



H. Infinite Maze

2 seconds, 256 megabytes

We've got a rectangular $n \times m$ -cell maze. Each cell is either passable, or is a wall (impassable). A little boy found the maze and cyclically tiled a plane with it so that the plane became an infinite maze. Now on this plane cell (x, y) is a wall if and only if cell $(x \bmod n, y \bmod m)$ is a wall.

In this problem $a \bmod b$ is a remainder of dividing number a by number b .

The little boy stood at some cell on the plane and he wondered whether he can walk infinitely far away from his starting position. From cell (x, y) he can go to one of the following cells: $(x, y - 1)$, $(x, y + 1)$, $(x - 1, y)$ and $(x + 1, y)$, provided that the cell he goes to is not a wall.

Input

The first line contains two space-separated integers n and m ($1 \leq n, m \leq 1500$) — the height and the width of the maze that the boy used to cyclically tile the plane.

Each of the next n lines contains m characters — the description of the labyrinth. Each character is either a "#", that marks a wall, a ".", that marks a passable cell, or an "S", that marks the little boy's starting point.

The starting point is a passable cell. It is guaranteed that character "S" occurs exactly once in the input.

Output

Print "Yes" (without the quotes), if the little boy can walk infinitely far from the starting point. Otherwise, print "No" (without the quotes).

input
5 4 ##. # ##S# #.. # #.# # #.. #
output
Yes

input
5 4 ##. # ##S# #.. # ..#. #.# #
output
No

In the first sample the little boy can go up for infinitely long as there is a "clear path" that goes vertically. He just needs to repeat the following steps infinitely: up, up, left, up, up, right, up.

In the second sample the vertical path is blocked. The path to the left doesn't work, too — the next "copy" of the maze traps the boy.

I. Feeding Chicken

1.5 seconds, 256 megabytes

Long is a huge fan of CFC (Codeforces Fried Chicken). But the price of CFC is increasing, so he decides to breed the chicken on his own farm.

His farm is presented by a rectangle grid with r rows and c columns. Some of these cells contain rice, others are empty. k chickens are living on his farm. **The number of chickens is not greater than the number of cells with rice on the farm.**

Long wants to give his chicken playgrounds by assigning these farm cells to his chickens. He would like to satisfy the following requirements:

- Each cell of the farm is assigned to **exactly one** chicken.
- Each chicken is assigned **at least one** cell.
- The set of cells assigned to every chicken forms a connected area. More precisely, if two cells (x, y) and (u, v) are assigned to the same chicken, this chicken is able to walk from (x, y) to (u, v) by passing only its cells and moving from each cell to another cell sharing a side.

Long also wants to prevent his chickens from fighting for food. Hence he wants the difference between the maximum and the minimum number of cells with rice assigned to a chicken to be as small as possible. Please help him.

Input

Each test contains multiple test cases. The first line contains the number of test cases T ($1 \leq T \leq 2 \cdot 10^4$). Description of the test cases follows.

The first line of each test case contains three integers r , c and k ($1 \leq r, c \leq 100, 1 \leq k \leq 62$), representing the size of Long's farm and the number of chickens Long has.

Each of the next r lines contains c characters, each is either "." or "R", representing an empty cell or a cell with rice. It is guaranteed that **the number of chickens is not greater than the number of cells with rice on the farm.**

It is guaranteed that the sum of $r \cdot c$ over all test cases does not exceed $2 \cdot 10^4$.

Output

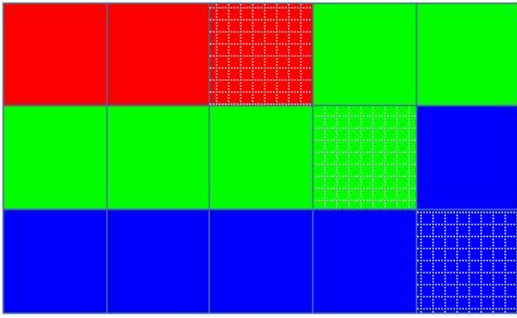
For each test case, print r lines with c characters on each line. Each character should be either a lowercase English character, an uppercase English character, or a digit. Two characters should be equal if and only if the two corresponding cells are assigned to the same chicken. **Uppercase and lowercase characters are considered different, so "A" and "a" belong to two different chickens.**

If there are multiple optimal answers, print any.

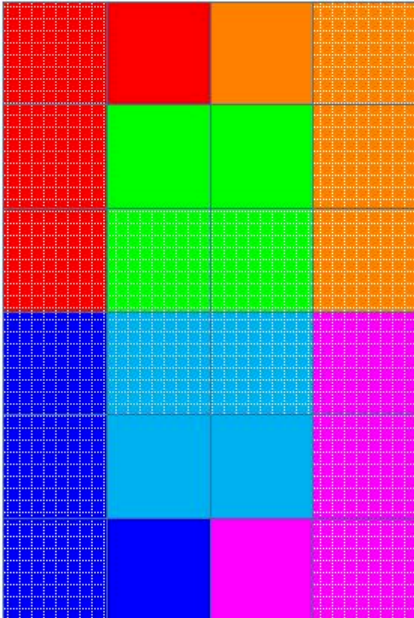
[illegible]

These pictures explain the sample output. Each color represents one chicken. Cells filled with patterns (not solid colors) contain rice.

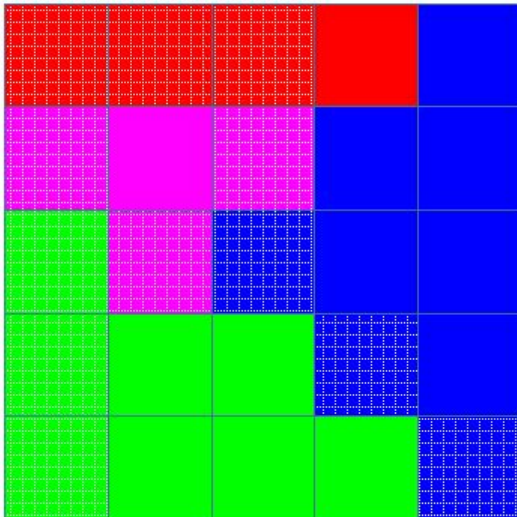
In the first test case, each chicken has one cell with rice. Hence, the difference between the maximum and the minimum number of cells with rice assigned to a chicken is 0.



In the second test case, there are 4 chickens with 3 cells of rice, and 2 chickens with 2 cells of rice. Hence, the difference between the maximum and the minimum number of cells with rice assigned to a chicken is $3 - 2 = 1$.



In the third test case, each chicken has 3 cells with rice.



In the last test case, since there are 62 chicken with exactly 62 cells of rice, each chicken must be assigned to exactly one cell. The sample output is one of the possible way.

J. Changing Volume

1 second, 256 megabytes

Bob watches TV every day. He always sets the volume of his TV to b . However, today he is angry to find out someone has changed the volume to a . Of course, Bob has a remote control that can change the volume.

There are six buttons (-5 , -2 , -1 , $+1$, $+2$, $+5$) on the control, which in one press can either increase or decrease the current volume by 1, 2, or 5. The volume can be arbitrarily large, but can never be negative. In other words, Bob cannot press the button if it causes the volume to be lower than 0.

As Bob is so angry, he wants to change the volume to b using as few button presses as possible. However, he forgets how to do such simple calculations, so he asks you for help. Write a program that given a and b , finds the minimum number of presses to change the TV volume from a to b .

Input

Each test contains multiple test cases. The first line contains the number of test cases T ($1 \leq T \leq 1\,000$). Then the descriptions of the test cases follow.

Each test case consists of one line containing two integers a and b ($0 \leq a, b \leq 10^9$) — the current volume and Bob's desired volume, respectively.

Output

For each test case, output a single integer — the minimum number of presses to change the TV volume from a to b . If Bob does not need to change the volume (i.e. $a = b$), then print 0.

input
3 4 0 5 14 3 9
output
2 3 2

In the first example, Bob can press the -2 button twice to reach 0. Note that Bob can not press -5 when the volume is 4 since it will make the volume negative.

In the second example, one of the optimal ways for Bob is to press the $+5$ twice, then press -1 once.

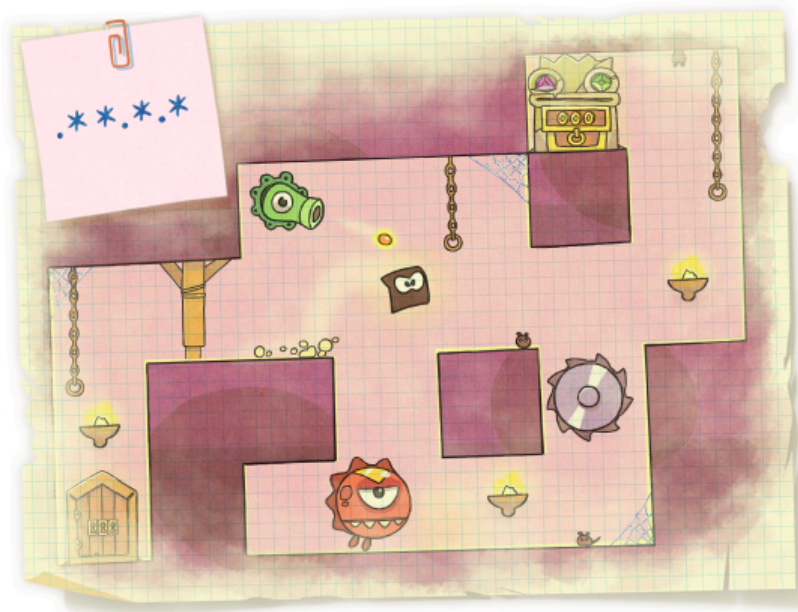
In the last example, Bob can press the $+5$ once, then press $+1$.

K. King of Thieves

1 second, 256 megabytes

In this problem you will meet the simplified model of game King of Thieves.

In a new ZeptoLab game called "King of Thieves" your aim is to reach a chest with gold by controlling your character, avoiding traps and obstacles on your way.



An interesting feature of the game is that you can design your own levels that will be available to other players. Let's consider the following simple design of a level.

A dungeon consists of n segments located at a same vertical level, each segment is either a platform that character can stand on, or a pit with a trap that makes player lose if he falls into it. All segments have the same length, platforms on the scheme of the level are represented as `'*'` and pits are represented as `'.'`.

One of things that affects speedrun characteristics of the level is a possibility to perform a series of consecutive jumps of the same length. More formally, when the character is on the platform number i_1 , he can make a sequence of jumps through the platforms $i_1 < i_2 < \dots < i_k$, if $i_2 - i_1 = i_3 - i_2 = \dots = i_k - i_{k-1}$. Of course, all segments $i_1, i_2, \dots, i_k$ should be exactly the platforms, not pits.

Let's call a level to be *good* if you can perform a sequence of **four** jumps of the same length or in the other words there must be a sequence $i_1, i_2, \dots, i_5$, consisting of **five** platforms so that the intervals between consecutive platforms are of the same length. Given the scheme of the level, check if it is good.

Input

The first line contains integer n ($1 \leq n \leq 100$) — the number of segments on the level.

Next line contains the scheme of the level represented as a string of n characters '*' and '.'.

Output

If the level is *good*, print the word "yes" (without the quotes), otherwise print the word "no" (without the quotes).

input
16 .*.*.*.*.*.*.*.*.
output
yes

input
11 .*.*.*.*.*.*.
output
no

In the first sample test you may perform a sequence of jumps through platforms 2, 5, 8, 11, 14.

L. Cashier

1 s., 256 MB

Student Arseny likes to plan his life for n days ahead. He visits a canteen every day and he has already decided what he will order in each of the following n days. Prices in the canteen do not change and that means Arseny will spend c_i rubles during the i -th day.

There are 1-ruble coins and 100-ruble notes in circulation. At this moment, Arseny has m coins and a sufficiently large amount of notes (you can assume that he has an infinite amount of them). Arseny loves modern technologies, so he uses his credit card everywhere except the canteen, but he has to pay in cash in the canteen because it does not accept cards.

Cashier always asks the student to pay change-free. However, it's not always possible, but Arseny tries to minimize the *dissatisfaction* of the cashier. Cashier's dissatisfaction for each of the days is determined by the total amount of notes and coins in the change. To be precise, if the cashier gives Arseny x notes and coins on the i -th day, his dissatisfaction for this day equals $x \cdot w_i$. Cashier always gives change using as little coins and notes as possible, he always has enough of them to be able to do this.

Arseny wants to pay in such a way that the total dissatisfaction of the cashier for n days would be as small as possible. Help him to find out how he needs to pay in each of the n days!

Note that Arseny always has enough money to pay, because he has an infinite amount of notes. Arseny can use notes and coins he received in change during any of the following days.

Input

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $0 \leq m \leq 10^9$) — the amount of days Arseny planned his actions for and the amount of coins he currently has.

The second line contains a sequence of integers $c_1, c_2, \dots, c_n$ ($1 \leq c_i \leq 10^5$) — the amounts of money in rubles which Arseny is going to spend for each of the following days.

The third line contains a sequence of integers $w_1, w_2, \dots, w_n$ ($1 \leq w_i \leq 10^5$) — the cashier's dissatisfaction coefficients for each of the following days.

Output

In the first line print one integer — minimum possible total dissatisfaction of the cashier.

Then print n lines, the i -th of them should contain two numbers — the amount of notes and the amount of coins which Arseny should use to pay in the canteen on the i -th day.

Of course, the total amount of money Arseny gives to the cashier in any of the days should be no less than the amount of money he has planned to spend. It also shouldn't exceed 10^6 rubles: Arseny never carries large sums of money with him.

If there are multiple answers, print any of them.

input
5 42 117 71 150 243 200 1 1 1 1 1
output
79 1 17 1 0 2 0 2 43 2 0

input
3 0 100 50 50 1 3 2
output
150 1 0 1 0 0 50

input
5 42 117 71 150 243 200 5 4 3 2 1
output
230 1 17 1 0 1 50 3 0 2 0

M. Lynyrd Skynyrd

2 seconds, 256 megabytes

Recently Lynyrd and Skynyrd went to a shop where Lynyrd bought a permutation p of length n , and Skynyrd bought an array a of length m , consisting of integers from 1 to n .

Lynyrd and Skynyrd became bored, so they asked you q queries, each of which has the following form: "does the subsegment of a from the l -th to the r -th positions, inclusive, have a subsequence that is a cyclic shift of p ?" Please answer the queries.

A *permutation* of length n is a sequence of n integers such that each integer from 1 to n appears exactly once in it.

A *cyclic shift* of a permutation $(p_1, p_2, \dots, p_n)$ is a permutation $(p_i, p_{i+1}, \dots, p_n, p_1, p_2, \dots, p_{i-1})$ for some i from 1 to n . For example, a permutation $(2, 1, 3)$ has three distinct cyclic shifts: $(2, 1, 3)$, $(1, 3, 2)$, $(3, 2, 1)$.

A *subsequence* of a subsegment of array a from the l -th to the r -th positions, inclusive, is a sequence $a_{i_1}, a_{i_2}, \dots, a_{i_k}$ for some $i_1, i_2, \dots, i_k$ such that $l \leq i_1 < i_2 < \dots < i_k \leq r$.

Input

The first line contains three integers n, m, q ($1 \leq n, m, q \leq 2 \cdot 10^5$) — the length of the permutation p , the length of the array a and the number of queries.

The next line contains n integers from 1 to n , where the i -th of them is the i -th element of the permutation. Each integer from 1 to n appears exactly once.

The next line contains m integers from 1 to n , the i -th of them is the i -th element of the array a .

The next q lines describe queries. The i -th of these lines contains two integers l_i and r_i ($1 \leq l_i \leq r_i \leq m$), meaning that the i -th query is about the subsegment of the array from the l_i -th to the r_i -th positions, inclusive.

Output

Print a single string of length q , consisting of 0 and 1, the digit on the i -th positions should be 1, if the subsegment of array a from the l_i -th to the r_i -th positions, inclusive, contains a subsequence that is a cyclic shift of p , and 0 otherwise.

input
3 6 3 2 1 3 1 2 3 1 2 3 1 5 2 6 3 5
output
110

input
2 4 3 2 1 1 1 2 2 1 2 2 3 3 4

output**010**

In the first example the segment from the 1-st to the 5-th positions is 1, 2, 3, 1, 2. There is a subsequence 1, 3, 2 that is a cyclic shift of the permutation. The subsegment from the 2-nd to the 6-th positions also contains a subsequence 2, 1, 3 that is equal to the permutation. The subsegment from the 3-rd to the 5-th positions is 3, 1, 2, there is only one subsequence of length 3 (3, 1, 2), but it is not a cyclic shift of the permutation.

In the second example the possible cyclic shifts are 1, 2 and 2, 1. The subsegment from the 1-st to the 2-nd positions is 1, 1, its subsequences are not cyclic shifts of the permutation. The subsegment from the 2-nd to the 3-rd positions is 1, 2, it coincides with the permutation. The subsegment from the 3 to the 4 positions is 2, 2, its subsequences are not cyclic shifts of the permutation.

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